

Scaffolded Tactivity for Solving Quadratic Equations Part 1

Solve the following Quadratic using the Square Root Method

$$(2y + 3)^2 - 9 = 0$$

$$(2y + 3)^2 = 9$$

$$\sqrt{(2y + 3)^2} = \pm\sqrt{9}$$

$$2y + 3 = \pm 3$$

$$2y + 3 = 3 \quad \text{or} \quad 2y + 3 = -3$$

$$2y = 0 \quad \text{or} \quad 2y = -6$$

$$y = 0 \quad \text{or} \quad y = -3$$

Solve the following Quadratic by using the Factoring Method

$$x(x - 8) + 12 = 0$$

$$x^2 - 8x + 12 = 0$$

$$(x - 6)(x - 2) = 0$$

By Zero-Product Rule we know that

$$x - 6 = 0 \text{ or } x - 2 = 0$$

$$x = 6 \text{ or } x = 2$$

Solve the following Quadratic by Completing the Square

$$x^2 - 6x - 13 = 0$$

$$x^2 - 6x + \underline{\hspace{1cm}} = 13 + \underline{\hspace{1cm}}$$

$$x^2 - 6x + 9 = 13 + 9$$

$$(x - 3)^2 = 22$$

$$\sqrt{(x - 3)^2} = \pm\sqrt{22}$$

$$x - 3 = \pm\sqrt{22}$$

$$x = 3 + \sqrt{22} \text{ or } x = 3 - \sqrt{22}$$

Solve the following Quadratic by using the Quadratic Formula

$$2x^2 - 3x - 1 = 0$$

Given the Quadratic Formula of $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ with $a = 2$; $b = -3$ and $c = -1$

$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(-1)}}{2(2)}$
$x = \frac{3 \pm \sqrt{9+8}}{4}$
$x = \frac{3 + \sqrt{17}}{4} \quad \text{or} \quad x = \frac{3 - \sqrt{17}}{4}$

Scaffolded Tactivity for Solving Quadratic Equations Part 2

Solve the quadratic equation $x^2 + 4x - 21 = 0$ by circling the next correct step in each row. In the third column explain why this is the correct step and what caused the error in the incorrect step.

Step 1	$x^2 + 4x + \underline{\hspace{1cm}} = -21 + \underline{\hspace{1cm}}$	$x^2 + 4x + \underline{\hspace{1cm}} = 21 + \underline{\hspace{1cm}}$	
Step 2	$x^2 + 4x + 4 = 21 + 4$	$x^2 + 4x + 16 = 21 + 16$	
Step 3	$(x + 2)^2 = 25$	$(x + 4)^2 = 25$	
Step 4	$\pm \sqrt{(x + 2)^2} = \sqrt{25}$	$\sqrt{(x + 2)^2} = \pm \sqrt{25}$	
Step 5	$x + 2 = \pm 5$	$x + 2 = 5$	
Step 6	$x + 2 = 5 \quad \text{or} \quad x + 2 = -5$	$x + 2 = 5 \quad \text{or} \quad x - 2 = -5$	
Step 7	$x = 3 \quad \text{or} \quad x = -3$	$x = 3 \quad \text{or} \quad x = -7$	

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Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by circling the next correct step in each row. In the third column explain why this is the correct step and what caused the error in the incorrect step.

Step 1	$(2x - 1)(x + 3) = 0$	$(2x + 1)(x - 3) = 0$	
Step 2	$2x + 1 = 0$ or $x - 3 = 0$	$2x - 1 = 0$ or $x + 3 = 0$	
Step 3	$x = \frac{1}{2}$ or $x = -3$	$x = -\frac{1}{2}$ or $x = 3$	

Scaffolded Tactivity for Solving Quadratic Equations Part 3

Find the real solutions, if any, of the equation $2x(x + 2) = 3$ by using the quadratic formula and filling in the boxes with the correct mathematical expression.

Step 1 Plug in values of a, b and c	$x = \frac{-\boxed{} \pm \sqrt{\boxed{}^2 - 4\boxed{}}}{2\boxed{}}$
Step 2	$x = \frac{-\boxed{} \pm \sqrt{\boxed{} + \boxed{}}}{\boxed{}}$
Step 3	$x = \frac{-\boxed{} \pm \sqrt{\boxed{}}}{\boxed{}}$
Step 4 Simplify	$x = \frac{-\boxed{} \pm \boxed{}\sqrt{\boxed{}}}{\boxed{}}$

Find the real solutions, if any, of the equation $x^2 + \frac{2}{3}x - \frac{1}{3} = 0$ by completing the square and then using the square root method. Fill in the boxes with the correct mathematical expressions as you proceed.

Step 1	$x^2 + \frac{2}{3}x + \boxed{} = \frac{1}{3} + \boxed{}$
Step 2	$(x + \boxed{})^2 = \boxed{}$
Step 3	$\sqrt{\boxed{}^2} = \pm \sqrt{\boxed{}}$
Step 4	$x = \boxed{} \text{ or } x = \boxed{}$

Scaffolded Tactivity for Solving Quadratic Equations Part 4

Find the real solutions, if any for the following equations by using any method you know.

1. $2y^2 - 50 = 0$

2. $(3z - 2)^2 = 4$

3. $x^2 - \frac{1}{2}x - \frac{3}{16} = 0$

4. $4t^2 + t + 1 = 0$

5. $25x^2 + 16 = 40x$

6. $x^2 + 6x + 1 = 0$